

GRADUATION PROJECT 2025 – 2026

Sportify: Cloud-Native Operations Platform

Multi-Sport Club Management System (MSCMS) via Microservices

Research Team:

M. Mohamed Ahmed, M. Abdellatief, E. Ahmed Elboghdady,
K. Mohamed, S. Abdelaziz, M. Salem

Supervised By:

Dr. Shaimaa Saber
Menoufia University | Faculty of Computers & Info

The Spotify Roadmap



Problem



Architecture



Implementation



Analytics



Future

Problem Definition: Fragmented Clubs

Fragmented Tooling

Club operations are scattered across legacy spreadsheets, paper medical records, and disconnected messaging apps, causing critical data siloing.

Lack of Intelligence

Decisions rely on coach intuition rather than data. Without a unified match engine, real-time tactical adjustments and predictive analytics remain impossible.

Mission Strategic Objectives

Unified Ecosystem: Consolidate user, squad, match, and medical data into a single role-aware cloud platform.

Match Engine: Deliver a broadcast-style live-match engine with an accurate timeline of tactical events.

Healthcare Lifecycle: Track every stage of player injuries, from diagnosis to passing the final fitness test.

Predictive Insights: Leverage Machine Learning to forecast match outcomes and evaluate player performance.

The Proposed Solution

Sportify replaces the patchwork approach with a **Cloud-Native Microservices Architecture**.

By leveraging an authoritative back-end and a modular front-end, we ensure every stakeholder—from the CEO to the physiotherapist—has a tailored, real-time dashboard powered by secure enterprise standards.



Role-Aware UI



Scalable Cloud

Microservices Orchestration

Our architecture consists of 7 isolated Spring Boot services coordinated via **Spring Cloud API Gateway**. JWT verification, async event processing through **Kafka**, and a dedicated **Python ML engine** ensure a robust, production-grade backbone.

Enterprise-Grade Tech Stack



Back-End

Java 21 / Spring Boot 3.5
Spring Cloud Gateway
PostgreSQL 14



Front-End

Next.js 16 (App Router)
React 19 & Tailwind v4
WebSockets for Real-time



Infrastructure

Keycloak 26 (Identity)
Apache Kafka (Eventing)
Docker Containerization

The Live-Match Logic Engine

Scheduled

Fixture ingestion from
football-data.org API.

Regulation

Real-time event tracking
(Goals, Cards, Subs).

Tie-Breaker

Extra Time & Penalty phase
logic for knockouts.

Persisted

Final results update league
tables & analytics.

Advanced Injury Management



Staged Recovery Workflow

Sportify implements a rigorous medical journey that acts as the single source of truth for player availability. When an injury is logged, it automatically triggers:

- ✓ Removal from match selection lists.
- ✓ Staged tracking: Diagnosis & Rehab.
- ✓ Passing a mandatory Fitness Test.

Data Intelligence & Match Predictions



Our **FastAPI ML Service** consumes historical data via Kafka to predict match outcomes and player ratings with high accuracy.

Real-time Tactical adjustments are supported by live form analysis, providing coaches with a **Competitive Edge**.

Future Expansion Roadmap



Wearable Integration

Ingesting biometric data for real-time load management.



Native Mobile

iOS/Android apps for scouts and medical staff.



Public Fan Portal

Extending read-only data for fan engagement and portals.



QUESTIONS & DISCUSSION

Thank You for Your Attention

Sportify MSCMS | graduation-defense-2026.fci.mu

✉ contact@sportify-project.com

🔗 github.com/sportify-microservices

Image Sources



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